

Knox, H. (2024). Doing place through data: Proliferation, profiling and the perils of portrayal in local climate action. *Big Data & Society*, 11(3).

<https://doi.org/10.1177/20539517241266404>

- **Categories:** Data, environment, social justice, community,
- **Summary:** “As climate and energy data increasingly demand that places become enrolled into environmental projects our findings suggest that data enables place to emerge as a ‘socio-technical potentiality’ an observation that has implications both for both engagement with, and the study of data and place.”

Munk, A. K., Jacomy, M., Ficozzi, M., & Jensen, T. E. (2024). Beyond artificial intelligence controversies: What are algorithms doing in the scientific literature?

Big Data & Society, 11(3). <https://doi.org/10.1177/20539517241255107>

- **Categories:** Semantic analysis, AI, controversy over AI
- **Method:** “Through a systematic quali-quantitative exploration of 235 co-word clusters and a subsequent structured search for 18 issue-specific queries, for which we devise a novel method with a custom-built datascape, we explore how algorithms have agency”
- **Argument:** Makes the argument that although expert analysis (versus popular/media portrayal of AI’s controversies) tend to focus on AI technologies in the context of specific problems. **However**, rather than focusing on the “controversiality of such technologies [like AI],” scientific literature has a proclivity for AI-driven solutions over AI-caused problems. The author argues that despite their attention to the specific workings of algorithms, scientific literature fails to take issue with AI.
- **Implication:** For me, the conclusion of the paper calls for a more “scientific” approach to AI that does not only focus on specific workings of algorithms but politicizes its general form through various approaches. Helpful quote:
 - “Thinking about AI as a broader political situation, we suggest, can be seen as a contribution to [Kitchin's \(2017\)](#) strategies for unpacking the full socio-technical assemblage of algorithms and examining how they do

work in the world. Many scholars, Kitchin included, have called for better algorithmic literacy to further public engagement with AI. On a basic level, this could mean acknowledging the presence of an algorithm in a system and building hypotheses about what its function might be ([Rieder, 2017](#)), whereas on a higher level it could mean the ability to assess and interact with algorithmic systems through informed decisions ([Klug et al., 2023](#); [Cotter, 2019](#)).”

- **Related:**

- Rieder B (2017) Scrutinizing an algorithmic technique: The Bayes classifier as interested reading of reality. *Information, Communication & Society*, 20(1): 100–117.
- Kitchin R (2017) Thinking critically about and researching algorithms. *Information, communication & society* 20(1): 14–29.

Tucker, A. (2024). Interoperable and standardized algorithmic images: The domestic war on drugs and mugshots within facial recognition technologies. *Big Data & Society*, 11(3). <https://doi.org/10.1177/20539517241274593>

- **Categories:** generative AI, War on Drugs, facial recognition technology, race and gender, representation, eugenics, marginalized groups
- **Method:**
 - Analyzes data practices and representations of mugshot databases and how FRT uses them
- **Main points:**
 - The article describes the coalescing history between facial recognition technology as a measure of digital policing and the gendered + racialized dynamics of mugshots in the War on Drugs.
 - The author examines a set of protocols related to official WoD mugshot datasets that created “interoperable” image-data; these image-data are meant to be “operationalized faster and in more frictionless manners than the generation of infrastructure before,” thus optimizing algorithmic policing and governance.

- The carceral logic of FRT & its data is replicated by generative AI systems like DALL-E and Midjourney, which are trained on facial data that are available and circulating online
- **Relevance:**
 - Has an interesting section in the conclusion where it talks about “the manipulation of prompts” allows us to ask such systems to “visualize their biases for us, to tell on themselves and reveal the logics within their deep learning and data practices”
 - Emphasizes the link between AI output and the (deep) racialized logics and representational protocols embedded within its training data
- **Related:**
 - Amaro R (2022) *The Black Technical Object: On Machine Learning and the Aspiration of Black Being*. Cambridge: MIT Press.

Kolkman, D., Bex, F., Narayan, N., & van der Put, M. (2024). *Justitia ex machina: The impact of an AI system on legal decision-making and discretionary authority*. *Big Data & Society*, 11(2). <https://doi.org/10.1177/20539517241255101>

- **Categories:** AI, Generative AI
- **Summary:** The authors of this article tackle the question of whether AI can effectively automate court processes. The conclusion is that they can't because they can't perform the kind of consistency and complex reasoning entailed in legal reasoning and the complex court process with a variety of materials & contingencies.
- **Method:** critical case study, they created an AI system as a part of their research and embedded that within courts / legal systems and compared whether AI was able to reach the same conclusion as paralegals. The result was negative.
- **Relevance:** it seems like this article is remotely related to something like legal language, but I don't think it is particularly relevant to our research, as it focuses more on the process of legal decision-making and less on the AI's ability to process and replicate - to that point, the article actually thinks an AI doesn't have the capacity to generate super meaningful results and legal decisions in its context.

Gillespie, T. (2024). Generative AI and the politics of visibility. *Big Data & Society*, 11(2). <https://doi.org/10.1177/20539517241252131>

- **Categories:** Large Language Models, AI bias, generative AI, Marginalized groups
- **Summary:** Asks the question of whether “cultural production” generated by AI can incorporate hard-fought representations of marginalized and non-normative communities. The conclusion is no. Through testing three generative AI tools, the author argues that AI almost always represents normative identities and narratives - even representations of marginalizes perspectives harbor deeper normative assumptions
- **Method:**
 - testing three gen AI tools with intentional prompts multiple times to generate text (stories and news articles) expose any biases / marginalizations;
 - Used “unmarked” prompts (i.e. prompts that didn’t reveal any gender/racial identity) like: *“Write a three-paragraph story in which two people meet, fall in love, and live happily ever after. Give the two characters names, jobs, and a favorite shared hobby.”*
 - Used some sort of text analysis / coding to analyze the prompts [e.g. if masculine and feminine pronouns were used, the marriage is straight] -> then used basic statistics to show the dominance of normative narratives
- **Relevance:**
 - The author’s analysis proceeds from more of a “cultural” and media perspective & explores the “representational harms” of AI - extending the question of what happens when cultural representations of marginalized communities are absent or stereotyped. He provides a helpful overview of this methodology in the first section
 - Introduces the idea of “markedness” - e.g. “wedding” is unmarked and normative and “gay wedding” is marked and non-normative - to show that unless specifically marked, AI produces unmarked (normative) images and narratives. This seems relevant to research on gender and sexuality

Stravato Emes, C. (2024). Is it about “them”? leveraging big data research to understand anti-immigrant discourse. *Big Data & Society*, 11(2).

<https://doi.org/10.1177/20539517241249432>

- **Categories:** social media, sentiment analysis, NLP, machine learning, data labeling
- **Summary:** The authors use machine learning to study social media attitude towards immigrants in Singapore.
- **Method:** supervised algorithmic extraction of patterns with human-based classification - first researcher defined patterns/classifications (annotating dataset), then training the classification algorithm. Used a dictionary method - I think this means mapping keywords with predetermined categories, assigning predetermined values to those categories, and reading them w/ NLP in an “automated textual analysis”
- **Relevance:** the results are mainly about Singaporean attitude on immigration, so not particularly relevant. But I appreciate the method and thought it was interesting - like how do you use NLP + human labeling to study certain attitudes.

Call, S. R., Jensen, J. T., & Barbour, J. B. (2024). Quantifying the self with others. *Big Data & Society*, 11(2). <https://doi.org/10.1177/20539517241247831>

- **Categories:** data, community, open-source
- **Summary:** It talks about how a community of “self-trackers” - people who use technology to track their bodily and health statuses - relate to and interact with data within a community of others.
- **Relevance:** not really, but it does include the body, which might be relevant

Peterson-Salahuddin, C. (2024). Repairing the harm: Toward an algorithmic reparations approach to hate speech content moderation. *Big Data & Society*, 11(2).

<https://doi.org/10.1177/20539517241245333>

- **Categories:** algorithm, machine learning, algorithmic bias, algorithmic reparation
- **Summary:** First proposes that using algorithms in detecting and limiting hate speech can be harmful as such algorithmic biases that can “silence or further harm marginalized users.” The author proposes the concept of “reparation,” a

less harmful way to detect and regulate hate speech using algorithms. Specifically, a reparative framework of content moderation might look at guiding the offender through a process of redress / repair that involves the community they offended

- **Relevance:** I appreciate the author offering a concrete solution to algorithmic bias that diverts its harmful effect into community-building strategies. The idea of reparation / repair also relates to Sedgwick's reparative reading.

Tracey, P., & Garcia, P. (2024). After automation: Homelessness prioritization algorithms and the future of care labor. *Big Data & Society*, 11(1).

<https://doi.org/10.1177/20539517241239043>

- **Categories:** algorithmic reparation, curation of data, algorithmic bias '
- **Summary:** The authors use qualitative methods to look at "the intertwining of data work and care labor of 15 care workers" in U.S. homeless service systems. Specifically the article deals with algorithmic prioritization in (homeless) service/care systems and how care workers grapple with the ethics of such a system and find workarounds - in a way that makes them key curators of the data.
- The article is based on the idea of "algorithmic reparation" - where "greater care is given to data work" beyond "surface-level diagnoses and band-aid solutions" within an opposite framework of algorithmic idealism. A part of *Big Data & Society's* [Algorithmic Reparations series](#).
- **Implication:** Reparation / reparative data work and data curation

Eglash, R., Robinson, K. P., Bennett, A., Robert, L., & Garvin, M. (2024). Computational reparations as generative justice: Decolonial transitions to unalienated circular value flow. *Big Data & Society*, 11(1).

<https://doi.org/10.1177/20539517231221732>

- **Categories,** reparation, computational reparation
- **Summary:** The article develops a framework where technology, instead of extracting resources and monopolizing value, can "act as restorative mechanisms for lost circular value flows" - basically on the possibility of technology facilitating alternative forms of labor and value creation.

- **Implication:** related to the idea of repair

Ng, L. H. X., Robertson, D. C., & Carley, K. M. (2024). Cyborgs for strategic communication on social media. *Big Data & Society*, 11(1).

<https://doi.org/10.1177/20539517241231275>

- **Categories:** automation, social media, NLP
- **Summary:** Use NLP, manual annotation, and network science to identify “Cyborg accounts” on social media platforms. Cyborg accounts are essentially bots that combine “manual online posting” and “automation” to evade the platform’s detection and push their strategic agenda.
- **Implication:** I think the NLP part is interesting & would be cool to look at how they annotate & analyze the text

Burkhardt, S., & Rieder, B. (2024). Foundation models are platform models: Prompting and the political economy of AI. *Big Data & Society*, 11(2).

<https://doi.org/10.1177/20539517241247839>

- **Categories:** generative AI, foundation models, critique of AI
- **Summary:** This article takes a detailed look at foundation models - models with extremely large datasets used in generative AI platforms like ChatGPT. It focuses on prompting as an algorithmic technique that makes these models programmable and accessible for different audiences. It ultimately argues that “foundation models and prompting further strengthen Big Tech’s dominance over the field of computing” and challenges regulatory response
- **Implication:** I thought this was a helpful critique of AI that looks at the specific form of it and how people engage with it, rather than focusing on the abstract idea of AI / algorithm. This article is also unique in terms of focusing on the production and economy of AI technology.

Stinson, C., & Vlaad, S. (2024). A feeling for the algorithm: Diversity, expertise, and artificial intelligence. *Big Data & Society*, 11(1).

<https://doi.org/10.1177/20539517231224247>

- **Categories:** AI bias, diversity, AI and marginalized groups
- **Summary:** Addresses the question of what diversity should look like as an adequate solution to the ethical problems of artificial intelligence. The authors

argue that the design and creation of artificial intelligence should incorporate workers of different “emergent expertise” - a phenomenon that is more networked than the skills of any individual alone.

- **Implication:** another of the various articles in this journal that focus on the interaction between AI and marginalized groups

Gourlet, P., Ricci, D., & Crépel, M. (2024). Reclaiming artificial intelligence accounts: A plea for a participatory turn in artificial intelligence inquiries. *Big Data & Society*, 11(2). <https://doi.org/10.1177/20539517241248093>

- **Categories:** AI controversy, participation in AI
- **Summary:** The article argues for and adopts a participatory method in problematizing AI through the situated and plural experiences of AI practitioners.
- **Implication:** This article is a part of Big Data & Society series on “[AI controversies](#)” - basically public discussion of AI, its history, and implications.

Helm, P., Lipp, B., & Pujadas, R. (2024). Generating reality and silencing debate: Synthetic data as discursive device. *Big Data & Society*, 11(2). <https://doi.org/10.1177/20539517241249447>

- **Categories:** generative AI, AI controversy, AI bias, platform economy
- **Summary:** The discussion centers around synthetic data and “data generation,” where certain proponents of AI argue that the use of AI-generated (rather than directly collected) data is ethically ok because training data is separated from real persons. The authors instead argue that “synthetic data” itself is political - it provides a sociotechnical imaginary that silences debates and a problematic solution to controversies over AI and big platforms.
- **Implication:** I appreciate that this article takes a critical stance and focuses on a specific thing - synthetic data. I also like how this article combines some technical details about the creation and use of synthetic data with larger socio political impacts.

Engdahl, I. (2024). Agreements ‘in the wild’: Standards and alignment in machine learning benchmark dataset construction. *Big Data & Society*, 11(2).

<https://doi.org/10.1177/20539517241242457>

- **Categories:** AI bias, data curation, data/database
- **Summary:** The article presents an ethnographic case study of the creation of a new “benchmark” dataset for machine learning tasks in a corporate-academic setting. Benchmarks include metrics, curated data, and task specifications that set the standard for a machine learning model. The article emphasizes the kind of “alignment” work that takes place during the creation of the dataset where different stakeholders navigate the creation of a standard that their model will be benchmarked against. The author adds to the argument that such datasets are socio-technical products and argues that they are far from capturing “unscripted activities” out happening in reality, but instead reflect the kind of intentional alignment work that precedes it.
- **Implication:** I thought this article provided a helpful entry point into something like “critical data and algorithm studies”

Wang, S., Cooper, N., & Eby, M. (2024). From human-centered to social-centered artificial intelligence: Assessing ChatGPT’s impact through disruptive events. *Big Data & Society*, 11(4). <https://doi.org/10.1177/20539517241290220>

- **Keywords:** generative AI, LLMs, ChatGPT, AI controversy
- **Summary:** This article critiques conventional evaluations of the biases, privacy issues, and performance of public LLMs as “human-centered” (especially in scientific ML research) - they “cast atomized individuals as the key recipients of technology’s benefits and detriments.” The authors instead come from a “social-centered framework” to evaluate the social impact of LLMs, where they “examine the distribution of benefits [and harms of LLM] in situated group and institutional contexts.”
- **Method:** the paper is largely explanatory / theoretical. It discusses the process of ChatGPT’s “quality control,” where OpenAI removes harmful content from the model through individual, “user-centered testing,” which neglects a more community-oriented approach. The authors then go through three case studies

from the domain of law, education, and work to illustrate the effect of LLMs in different institutional contexts.

- **Implication:** I find the article helpful for introducing the difference between individual-centered and social-centered analysis of AI. However, it mostly focuses on institutional ideas of , while neglecting other forms of social/community groupings.

FACCT 2023 - [full list of articles](#)

“Examining risks of racial biases in NLP tools for child protective services.” FaCCT 2023. <https://dl.acm.org/doi/10.1145/3593013.3594094>

- **Categories:** AI bias, AI controversy, race and racism, NLP
- **Summary:** Looks at potential liabilities that might occur when AI is used on Child Protective Services data (case notes that CPS workers write on families and children). Specifically, they examine **word statistics** (word2vec), **algorithmic fairness in risk prediction** (no significant unfairness), **named entity recognition** (significantly unfair - where various models can better recognize/identify existing family members for White children than Black children in the dataset), and **coreference resolution** (didn't read far enough to know this). They find particular algorithmic unfairness in NER models.
- **Implication:** I appreciate that this studies AI in a real context with actual (rather than curated) data. I think the approach might be similar to our research? I also appreciate that they highlight several ways to study bias within an AI model.

“Which Stereotypes Are Moderated and Under-Moderated in Search Engine Autocompletion?” FaCCT 2023. <https://dl.acm.org/doi/10.1145/3593013.3594062>

- **Categories:** gender/sexuality, stereotypes, large language models, NLP,
- **Summary:** Studies the regulation of racist / sexist and otherwise discriminatory stereotypes in autocomplete features in Google, Yahoo, and Duckduckgo. The researchers prompted each search engine with “Why are [...] so [...]” using Python. Research found that sexual orientations are the most regulated - without any negative stereotypes showing up in autocomplete. Gender/identity

on the other hand is less regulated, with misogynistic stereotypes surfacing. No negative autocomplete results for “trans” or “transgender”

- **Implication:** I thought this was an interesting study intersecting AI / predictive models and gender.

“Multi-dimensional Discrimination in Law and Machine Learning -

A Comparative Overview” <https://dl.acm.org/doi/pdf/10.1145/3593013.3593979>

- **Categories:** discrimination, AI bias, AI controversy, law/legal, intersectionality
- **Summary:** This article raises the question of how to assess AI fairness in terms of multi-dimensionality / intersectionality rather than focusing on one characteristic/attribute, such as race, gender, or age. The article goes on to compare approaches to multi-dimensional discrimination in law and in machine learning, particularly exploring “additive and sequential discrimination.” The article also discusses how definitions of intersectionality might be transferred to improve fairness in machine learning.
- **Implication:** The legal aspect seems to dovetail with our research and it would be interesting to look at how the idea of intersectionality manifests itself in ML.

Week 3-4

FACCT 2022 - [full list of articles](#)

“Theories of ‘Gender’ in NLP Bias Research.” Hannah Devinney, Jenny Björklund, Henrik Björklund. <https://dl.acm.org/doi/10.1145/3531146.3534627>

- **Categories:** discrimination, AI bias, AI controversy, gender, non-binary, transgender, NLP
- **Summary:** The researchers survey 200 articles discussing NLP and gender to see how they theorize “gender,” a term that has not been well-defined within the field. They reach the conclusion that “” Almost none use a model of gender that is intersectional or inclusive of nonbinary genders; and **many conflate sex characteristics, social gender, and linguistic gender in ways that disregard the existence and experience of trans, nonbinary, and intersex people.**”
Though many articles recognize the making of gender is complicated, their practice/methods do not actually put this into practice.
 - They also provide recommendations for integrating bias research in NLP with Gender Studies:
- **Method:** surveys 200 articles discussing NLP and gender to see how they theorize “gender,” a term that has not been well-defined within the field.
- **Implication:** beyond the salient conclusion of the paper, I appreciate a section of the Discussion that focuses on “cisnormative analyses” in papers that they find problematic. “Cisnormative analysis” can include something like - researchers understanding the phrase “he gave birth,” which is statistically uncommon, as “biologically inaccurate” or “semantically incorrect” (see page 2087). I also appreciate their call for a “trans-inclusive methodology”

-----Relevant articles from this article-----

“Defining and Evaluating Fair Natural Language Generation”

<https://aclanthology.org/2020.winlp-1.27/>

- **Summary:** About using matrix / vector to assess gender fairness in natural language generation models

- **Implication:** Similar to our research in terms of looking at generated outputs, but the method they use is very technical - their goal here is to introduce a general model for assessing gender fairness in natural language generation outputs.

Yang Trista Cao and Hal Daumé III. 2020. “Toward Gender-Inclusive Coreference Resolution.” In Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics.

<https://aclanthology.org/2020.acl-main.418/>

- **Summary:** This article argues that current NLP systems fail to recognize and therefore excludes/fails to represent nonbinary gender identities. The article uses “coreference resolution” (“determining which textual references resolve to the same real-world entity”) to determine how much an NLP system is able to recognize nonbinary gender identities. Their research finds that NLP systems work less well with recognizing nonbinary gender because they rely heavily on referential gender (pronouns), which, the researchers argue, show that NLP systems “rely heavily on gender-based inferences” in parsing the meaning and inferences of a sentence, hence leading to exclusion of nonbinary gender identities.
- **Method:** 1) curated their own “gender neutral” dataset by using they/xe/ze pronouns, replacing names with initials, replacing gendered nouns with gender-indefinite variants, and removing term of address. 2) another dataset that consists of wikipedia articles about individuals who use nonbinary pronouns, articles from LGBTQ articles, and AO3 fan-fiction. Then analyzed the accuracy of coreference resolution in different NLP systems.
- **Implication:** I think this is one of the few articles that actually addresses nonbinary gender identities and experiences. It also incorporates this approach in a practical way - basically asking the question of what it would be like and what is needed for current NLP technology to process nonbinary / noncisgender identities.

“Towards Intersectional Feminist and Participatory ML: A Case Study in Supporting Femicide Counterdata Collection”

<https://dl.acm.org/doi/10.1145/3531146.3533132>

- **Keywords:** data collection, data curation, machine learning
- **Summary:** this article discusses a case study where participatory methods were used from the beginning to the end of a machine learning project cycle in the context of research + activism regarding femicide.
- **Method:** in their case study, they essentially developed a ML algorithm that could filter news articles, pick out the ones about femicide using a probability threshold, and send them to data activists who collect these data. The algorithm is based on participatory input from activists against femicide, where they collected and annotated two datasets of around 400 articles.
- **Implication:** I appreciate that this article talks about self-curated sets of data, annotation, and the development of a model. The more detailed methods they recount might be helpful.

Gauri Kambhatla, Ian Stewart, and Rada Mihalcea. 2022. “Surfacing Racial Stereotypes through Identity Portrayal.” <https://doi.org/10.1145/3531146.3533217>

- **Keywords:** Bias, stereotype, machine learning, NLP, race, racial stereotype
- **Summary:** This article conducts an experiment where a group of (white or Black) individuals write as themselves vs. write as someone from the other race. It then uses NLP to study and compare the language used by participants as themselves and as a false identity. In their methodology, if the machine is able to distinguish between the participants' real and portrayed racial identity, it means it has successfully identified the presence of racial stereotypes (the method is a bit more complicated than I describe as they do certain analyses to make sure that the difference in identity computed is not due to some factor other than racial stereotypes).
- **Implication:** The author argues that it is possible to “computationally surface racial stereotypes in texts by identifying linguistic features associated with differences in racial identity portrayal.” I find this intriguing and the method is

interesting and possibly relevant because it is about analyzing bias using NLP/machine learning.

“The Death of the Legal Subject: How Predictive Algorithms Are (Re)constructing Legal Subjectivity.” <https://dl.acm.org/doi/10.1145/3531146.3533134> Katrina Geddes (there was only an abstract for this)

- **Keywords:** law, algorithm and law, algorithmic bias
- **Summary:** The author compares the construction of legal subjectivity and subjectivity formed within the predictive mechanisms of informational capitalism. The author argues that “the introduction of predictive algorithms in judicial decision-making” and the production of legal subjects through algorithm, discount “qualitative knowledge about the legal subject’s intentions, motivations, and moral capabilities.”
- **Implication:** This is one of the few articles about law and algorithm that focuses on subjecthood. Although I don’t think it goes into the details of the use of algorithms in legal settings and legal language, it makes me wonder how the idea of subjecthood is also changed when we introduce trans/nonbinary into the intersection between law and algorithm.

FACCT 2021

Kendra Albert and Maggie Delano. 2021. This Whole Thing Smacks of Gender: Algorithmic Exclusion in Bioimpedance-based Body Composition Analysis. In Proceedings of the 2021 ACM Conference on Fairness, Accountability, and Transparency (FACCT '21). Association for Computing Machinery, New York, NY, USA, 342–352. <https://doi.org/10.1145/3442188.3445898>

- **Key words:** data collection and curation, sex/gender
- **Summary:** looks at the problem of smart weight scales to take up the issue of cisnormativity and nonbinary exclusion in personal health tech. Also provide

recommendations as to how to approach creating more inclusive technologies even while still relying on exclusionary data.

- **Implications:** it advocates for including trans perspectives and diagnoses nonbinary exclusion in everyday technology and their use of data/algorithm, but mostly in a more HCI fashion rather than generative AI or NLP.

Other articles I found while doing research

“The Misgendering Machines: Trans/HCI Implications of Automatic Gender Recognition” Proceedings of the ACM on Human-Computer Interaction, Volume 2, Issue CSCW. Os Keyes

- **Key words:** transgender (trans), HCI, automated gender recognition
- **Summary:** looks higher-level at academic/scientific articles on automated gender recognition and argues that they construe gender as binary, immutable, physiological, and *assigned*. Then argues that automated gender recognition is fundamentally at odds with a “trans-inclusive system” or perspective. However, the article also provides recommendations for remediating these problems, such as incorporating self-identification in the development of automated gender recognition technology.
- **Implication:** Although this article is not exactly related to our research, I thought the use of transgender/transness and using it to approach a specific technology is interesting - it argues how one technology is “ideologically” or just inherently at odds with something like transness. The author works on trans studies and AI and has other interesting work - their website <https://ironholds.org/>

“How We've Taught Algorithms to See Identity: Constructing Race and Gender in Image Databases for Facial Analysis.” Scheuerman et al.

<https://dl.acm.org/doi/10.1145/3392866>

[When do Word Embeddings Accurately Reflect Surveys on our Beliefs About People?](<https://aclanthology.org/2020.acl-main.405>) (Joseph & Morgan, ACL 2020)

- **Key words:** computational linguistics
- **Summary:** this article talks about word embeddings and how, and how much, they mirror human biases. After analyzing publicly available word embeddings for stereotypes, the conclusion reached is that these embeddings mirror real life stereotypes particularly in relation to gender and also to race, but to a lesser extent.
- **Implication:** this seems like an important article in computational linguistics that discusses the existence and measurement of bias in word embeddings/words as vectors.

Big Data & Society 2023

Wang, S., & Spronk, R. (2023). "Big data see through you": Sexual identifications in an age of algorithmic recommendation. *Big Data & Society*, 10(2).

<https://doi.org/10.1177/20539517231215358>

- **Keywords:** algorithm, social media, gender, sexuality, sexual identification
- **Summary:** this article argues that algorithmic recommendation on social media reshapes conceptions of gender/sexuality as simply a box that users select when registering for the service, but constructed through complex algorithmic networks and user engagement, leading to sexual orientations and desires being more fluid and multiple across time.
- **Method:** the authors study this phenomenon particularly on social media in China and interview queer and non-heterosexual individuals who use these social media to study their experience.
- **Implication:** I think this paper asks the interesting question of how can gender/sexuality be understood algorithmically or by an algorithm? Whereas other articles try to understand this by studying bias in the curation or data or the generation of content through algorithm, this article goes about it in a more social science/field work-like perspective.

Deitz, S. (2023). Outlier bias: AI classification of curb ramps, outliers, and context. *Big Data & Society*, 10(2). <https://doi.org/10.1177/20539517231203669>

- **Keywords:** algorithmic bias, AI bias, AI controversy, machine learning
- **Summary:** This article studies self-driving cars and the idea of data-infused smart-cities to make the point that algorithms are biased specifically because of their inability to deal with “**context, nuance, and outliers**”
- **Method:** the author uses aerial imagery and geographic data to create a data set of curb ramps and see if algorithms can predict where the ramps are.
- **Implication:** I think the claim that machine learning models “gravitate toward the mean” and therefore ignores and sidelines “context, nuance, and outliers” is an important one in thinking about LLMs as they deal with disruptions and nuances in language related to gender.

Hagen, S., & de Zeeuw, D. (2023). Based and confused: Tracing the political connotations of a memetic phrase across the web. *Big Data & Society*, 10(1). <https://doi.org/10.1177/20539517231163175>

- **Keywords:** social media, NLP, machine learning, data labeling, sentiment analysis
- **Summary:** The authors look at the transformation of the usage of a memed word in far-right contexts and platforms – the word is “based.” They employ quali-quantitative methods to trace the changing connotations and contexts of the word and conclude that far-right memes and hate speech - rather than carrying a singular problematic message - has “varied meanings and subcultural functions” in different contexts.
- **Method:** they conduct large-scale data analysis to trace the word’s meaning across time. First collecting all instances of “based” across three different platforms, then using “co-word” analysis to identify the words most commonly associated with “based,” then identifying the time periods on different platforms where the word received the most popularity/attention.
- **Implication:** I thought the method was particularly relevant because it combines pretty straightforward data extraction with more qualitative analysis. After using algorithmic methods to identify key nodes in the development of “based,” they look at the context and meaning of each instance. I think this combination of first using quantitative and then qualitative methods might be helpful.

Big Data & Society 2022

Nee, J., Smith, G. M., Sheares, A., & Rustagi, I. (2022). Linguistic justice as a framework for designing, developing, and managing natural language processing tools. *Big Data & Society*, 9(1). <https://doi.org/10.1177/20539517221090930>

- **Key words:** bias, NLP, stereotype, big data, linguistics, computational linguistics
- **Summary:** In this article, the authors argue that NLP tools can benefit from the perspective of linguistic justice, where it is analyzed from the perspective of minoritized language users - or generally marginalized users and how they interact with and access NLP. In particular, the authors emphasize the potential harms of NLP from a linguistic perspective, such as “failing to provide access to information, services or opportunities to users' language of choice, underperforming for certain linguistic groups, or advancing harmful stereotypes.” They argue that stereotypes arise from linguistic profiling, where not just the literal content of language, but the information surrounding it, contribute to certain stereotypes and assumptions within the NLP model.
- **Implication:** I don't think this article is particularly relevant but it seems like an earlier example from the more recent years that discusses bias in NLP technology from the perspective of language. I think it brings in linguistic terms to help understand bias in NLP/

Brooker, P. (2022). Computational ethnography: A view from sociology. *Big Data & Society*, 9(1). <https://doi.org/10.1177/20539517211069892>

- **Key words:** computational tools, anthropology
- **Summary:** This is a summary article that discusses computational ethnography, which uses computational tools for anthropological inquiry. The article identifies tension between “the context-specific and inherently-amorphous nature of (digital ethnographic studies” and the need for more structure/prescription for this kind of research.
- **Implications:** I initially thought this article would be helpful in outlining computational tools used for ethnographic/anthropological studies, but it

doesn't really do that. It's more of a musing on computational methods in general and the demands of the discipline of anthropology. But it does mention things like "social media analytics, data science, machine learning, natural language processing, corpus linguistics, etc.," which does make me wonder if our research might be related to any of these.

Liu, J. (2022). Social data governance: Towards a definition and model. Big Data & Society, 9(2). <https://doi.org/10.1177/20539517221111352>

- **Key words:** data governance
- **Summary:** This article thinks through the question of what are the "governing effects of data, dataism, and datafication," and their emergence within society. Rather than looking simply at how data governance is done at the organizational level, and how data is used to manage countries and populations, the article focuses more broadly on the macro - significance of governing with data. As the author writes, "the terms 'data' and 'governance' both entail the societal dynamics that underpin various dataism, datafication, and governance issues."
- **Implication:** I didn't finish this article because the author ends up arguing that "national culture" has an impact on social data governance. It's a frustratingly racist claim that recenters Western centrality.

Tacheva, Z. (2022). Taking a critical look at the critical turn in data science: From "data feminism" to transnational feminist data science. Big Data & Society, 9(2). <https://doi.org/10.1177/20539517221112901>

- **Keywords:** transnational feminism, data feminism, algorithmic oppression, data science
- **Summary:** the article uses feminist theory and the debates within it to decipher trends in feminist practices in data science. Tacheva critiques essentialist feminist practices and advocates for the recognition of interconnected struggles and communities within the work of data science. Also generally calls for moving away from the view that data / science is devoid of ideology and asks

data scientists to evaluate their own stance in relation to feminism and oppression.

Borch, C., & Hee Min, B. (2022). Toward a sociology of machine learning explainability: Human-machine interaction in deep neural network-based automated trading. *Big Data & Society*, 9(2).

<https://doi.org/10.1177/20539517221111361>

- **Keywords:** explainability
- **Summary:** This article uses a case study of “a trading firm’s quest for explaining its deep neural network systems actionable predictions” to explore what is explainability - how the opaque and complicated concepts of AI’s decision-making (e.g. deep neural networks) can be explained. They argue that the idea of “machine learning explainability” exemplifies “human-machine companionship.” The authors see this article as a “sociology of ML explainability,” where they understand “explainability” as an empirical object and use it to study the interaction between humans and machines.

Liminga, A., & Lindgren, S. (2024). Mapping the discursive landscape of data activism: Articulations and actors in an emerging movement. *Big Data & Society*, 11(3). <https://doi.org/10.1177/20539517241266416>

- **Keywords:** mixed-methods approach, computational analysis, discourse analysis
- **Summary:** this article uses computational methods to study the emergence of datafication discourse on Twitter. The authors used a “co-occurrence” method to track the frequency of how often two hashtags occur together. Then they mapped out the proximities among hashtags using a networked map/graph. Then they proceeded to analyze each cluster, like which were the most frequent hashtags in a certain cluster, etc.
- **Implication:** not much, but I thought the method might be relevant

Carlsen, H. B., & Ralund, S. (2022). Computational grounded theory revisited: From computer-led to computer-assisted text analysis. *Big Data & Society*, 9(1).

<https://doi.org/10.1177/20539517221080146>

- **Keywords:** text analysis, computational grounded theory
- **Summary:** The article compares approaches in text analysis in sociological / anthropological studies. The article distinguishes between two kinds of studies: supervised and unsupervised. Supervised means the human coder is inputting categories into the machine, while unsupervised text classification means “inductively deriving categories from the data, allowing for the polysemy of words and the heteroglossia of documents, assuming the relationality of meaning.” The article also introduces “computational grounded theory,” where theory/interpretation is based on computational analysis, which is perceived to remove human bias in analyzing and categorizing data. However, the article then argues against the claim of unbiased for unsupervised analysis. To prove their point, the authors conduct an experiment using popular “topic models” to show how they are unable to discover known planted topics.

Cultural Analytics

Emad Mohamed, Raheem Sarwar. "Exploring Gender Differences in Fatwa through Machine Learning." Vol. 9, Issue 3, 2024. Cultural Analytics.

<https://culturalanalytics.org/article/116368-exploring-gender-differences-in-fatwa-through-machine-learning>

- **Key words:** gender difference, online activity, data curation,
- **Summary:** The article uses machine learning to process and study a dataset consisting of question-answer pairs from a religious question-answering platform. produced in the context of Islamic "fatwa" (where people's questions are answered by a religious/legal figure according to Islamic law). The article studies the dataset uses NLP and machine learning to understand the difference between questions posed by women versus men, and they see if the algorithm can predict the gender of the person who asks a given question. They also consider the patterns behind popular answers. The results are presented as lists of questions, words, and topics that are most used by men/women.
- **Method:**
 - Pertaining to gender, the authors use "topic modeling" to find the themes of the questions raised by men and women and the "**odds ratios**" to determine which themes are more female and which tend to be more male. They try to predict the gender of a question using ML through "automatic machine learning," "text classification," and identifying lexical items that are more associated with men/women.
- **Relevance:** I thought this article was interesting because it uses machine learning to identify and classify gender in correlation with themes across a large amount of data. It's a bit disappointing that it doesn't address queer/non-binary perspectives. While reading this, I was curious about what if there are some questions that the algorithm fails to classify as male / female, or they have an assigned value that is in between the threshold for being identified as male/female? Instead of removing them from the algorithm, what could the researchers have done with it?

Quantifying Women's Marginalisation in Ibero-American Film Culture During the First Half of the Twentieth Century: A Network-Science Proposal. Vol. 9, Issue 4, 2024.

<https://culturalanalytics.org/article/118589-quantifying-women-s-marginalisation-in-ibero-american-film-culture-during-the-first-half-of-the-twentieth-century-a-network-science-proposal>

- **Key words:** gender difference, women, social network analysis
- **Summary:** The study uses social network analysis to study women's roles in Ibero-American film during the early 20th century. The network analysis shows that women existed in the outer margins of the empirical network, which leads the study to conclude both marginalization and a lack of evidence of the participation of women in the public sphere.
- **Method:**
 - They first reconstruct the social network of the Ibero-American film industry with a network of interactions between people involved in it.
 - Then, proposing a k-core decomposition and a multi-layered analysis.
[not sure what this is but some kind of social network analysis method]
- **Relevance:** I found that this article operates on a pretty binary definition of gender. It also doesn't really provide a concrete definition of gender as it relates to the method that it is working with (except assuming a very binary or identitarian one). I feel like maybe this is something missing from most scholarships that uses quantitative methods / data to study gender differences and inequality.

Revisiting Weimar Film Reviewers' Sentiments: Integrating Lexicon-Based Sentiment Analysis with Large Language Models. Vol. 9, Issue 4, 2024.

<https://culturalanalytics.org/article/118497-revisiting-weimar-film-reviewers-sentiments-integrating-lexicon-based-sentiment-analysis-with-large-language-models>

- **Keywords:** sentiment analysis, large language models, ChatGPT, lexicon based sentiment analysis
- **Summary:** The study uses ChatGPT and lexicon-based sentiment analysis (with Python's NLTK library and its VADER lexicon) to conduct sentiment analyses of historical film reviews to see how well they can accurately grasp nuanced sentiments. They found that ChatGPT was adept at dealing with nuanced language but could not be controlled and implemented in a consistent and reproducible way. Therefore, they advocated for their method, which first prompts ChatGPT to undertake a "verbose sentiment analysis," where the model explains its judgment of the film reviews. Then, it uses lexicon-based sentiment analysis on the response produced by ChatGPT to produce sentiment analysis results that have higher accuracy even when analyzing nuanced texts like reviews.
- **Methods:** Using large language models and lexicon-based sentiment analysis (using pre-built dictionaries containing words and their associated sentiment scores).
- **Relevance:** This was an interesting article. Although I thought the method was a little sketchy at first, I am surprised that it produced such an accurate result even when reading historical text. However, I do find the method a little simple, as the sentiments analyzed were basically "positive," "mixed," "negative," so it's hard to say how "nuanced" this is in terms of actually engaging with things like film reviews.

Torres-Cacoullos, Anna, and Elizaveta Senatorova. "A Digital Archaeology of Early Hispanic Film Culture: Film Magazines and the Male Fan Reader." *Journal of Cultural Analytics*, vol. 9, no. 4, July 2024, <https://doi.org/10.22148/001c.118152>.

- **Key words:** digital humanities, quantitative methods, gender,
- **Summary:** this study uses a combination of digital humanitie methods as well as close reading to study male fans in early hispanic film culture. It looks at archives of fan mails and a "reader star-search photo contest." The authors use digital methods to represent their archive in ways that create various information - such as what male fans were interested in, which celebrities they liked, how they see themselves, and their geographic and demographic information. It also concludes that while women have often been assumed to be the consumer of films and film magazines, men were actually more dominant in early film culture.
- **Methods:** The article uses a variety of methods to study digitized versions of the archive they were looking at - the methods include text mining, topic modeling, data processing, data visualization, GIS mapping, and network analysis using a co-occurrence matrix.
- **Relevance:** I thought the method of combining digital methods (they categorize this as "distant reading") and close reading methods particularly relevant; for example the authors would close read particular fan letters and use keyword search to find relevant data across all their archives.

Zaki, Mai, and Emad Mohamed. "Two Translations of Mahfouz's *Awlad Haratina* (*Children of Our Alley*): A Computational-Stylistic Analysis." *Journal of Cultural Analytics*, vol. 9, no. 3, June 2024, <https://doi.org/10.22148/001c.116239>.

- **Keywords:** translation, style, computational analysis, distant reading, machine learning
- **Summary:** The study uses computational methods to compare two translations of a canonical modern Arabic novel. The results show distinct patterns in "richness, sentence structure, readability level, stylometric analysis, and lexical choices." The study sets out to propose using computational methods and machine learning to identify "linguistic features which can reveal the translator's

thumbprints” and habits that are otherwise undetectable with a close reading method. It makes several observations about each text and attributes them to the contexts of their publication.

- **Methods:** They focus on extracting metrics such as vocabulary richness (using type-token ratio), sentence length, lexical bundles (aka sentences, extracted using Sketch Engine), readability (also using existing readability algorithms), stylometric analysis using Stylo (which uses n-grams to position the two texts in relation to each other in a 2D vector, or something long the lines of this...). They also use supervised machine learning, of the top 100 words in each text, to predict whether a specific piece of text belongs to a specific translator, and conclude that machine learning can do this relatively accurately. Finally, they use collocation patterns to study the conceptual patterns / meanings behind each book.
 - Screenshot of methods from the article:
 - Lexical variety (type-token ratio, lemma, hapax legomena)
 - Syntactic / narrative style (sentence length and lexical bundles)
 - Readability grade (level of comprehension difficulty)
 - Stylometric analysis (bootstrap consensus trees, principal components analysis, multidimensional scaling and cluster analysis)
 - Translator thumbprints (lexical choices and collocation)
- **Relevance:** I felt this article makes a compelling case for using machine learning / computational methods to detect unconscious linguistic patterns that are hard to detect without such methods. Although they attribute most of the differences that machine detects to simply contextual differences and motivations behind each translation, I am wondering if such methods can also be used to say

something literary or even specifically ideological about these texts - as well as how these differences detected actually influence how people read and understand these texts.

Lit, Cornelis van, and Dirk Roorda. "Neither Corpus Nor Edition: Building a Pipeline to Make Data Analysis Possible on Medieval Arabic Commentary Traditions."

Journal of Cultural Analytics, vol. 9, no. 3, June 2024,

<https://doi.org/10.22148/001c.116372>.

- **Keywords:** text analysis, computational analysis
- **Summary:** this article discusses building a suite of tools in Python to proficiently analyze text reuse and intertextuality. They talk about storing these specific sets of medieval Arabic texts within a data structure that mimics the intertextuality of the texts and how they perform textual analysis on them.
- **Relevance:** I didn't see a lot of relevance because this was mostly about preparing a specific set of manuscripts with an inherent system of intertextuality and a lot of it is about converting PDF to textual data, but they do use Python, which might be relevant.

Ryad, Umar, and Emad Mohamed. "A Topic Modelling of Muslim Religious Reform in the Colonial Age: A Computational and Digital Study of *al-Manār* (1898-1935)."

Journal of Cultural Analytics, vol. 9, no. 3, June 2024,

<https://doi.org/10.22148/001c.116225>.

- **Keywords:** text analysis, computational analysis, topic modeling, NLP, distant reading
- **Summary:** the article uses computational methods such as topic modeling, morphological processing to examine the co-occurrences of the topics and lexemes in a corpus of writing on Muslim reform in early 20th century Cairo. The article seeks to track, quantify, and explain the religious reformist ideas within the corpus. They conclude that their topic modeling method can accurately identify topics that can be mapped onto accurate historical developments affecting what people talked about at the time.

- **Method:** the authors automatically cluster the data (the corpus they study) into topics and narrow down the scope to look at each topic. Specifically, they use “probabilistic topic modeling.” The modeling method uses “co-occurrence statistics” and “the principles of distributional semantics” to automatically generate the most relevant words/themes, as well as the highest frequency words that cluster around it. They use a toolkit called Mallet to do this. The authors then use the topics generated to talk about what was happening historically and how that may have effected the discourses in the corpus.
- **Relevance:** I’m particularly interested in the “probabilistic topic modeling” that they used, which sounds like it automatically generated topics based on co-occurrence. I wonder how far this is from a generative model. The authors use the topics generated as entry points for a more historical analysis, but I am also wondering what else can the topics generated say about the texts they come from - e.g. sentiment, attitude, rhetoric, etc.

Karmanov, Fedor, and Joshua Kotin. “A Counterfactual Canon.” *Journal of Cultural Analytics*, vol. 9, no. 2, May 2024, <https://doi.org/10.22148/001c.116915>.

- **Keywords:** gender, data, data analysis
- **Summary:** The authors look at gender demographic data from the archive of the *Shakespeare and Company Project*. By analyzing the gender makeup of the borrowers of a book and the gender of the book author, they observe that that the reading practices of men determined the canon of female authors, and that the reading practices of women might reveal a counterfactual canon of modernism
- **Method:** just pretty standard data analysis
- **Relevance:** Although the definition of gender is pretty binary here (since the data they used did not really accommodate for queer/nonbinary gender), the authors do acknowledge the limitations of their method. This makes me wonder whether computational methods can be used to read for themes (i.e. queerness or transness) that are inherently limited by the dataset (i.e. gender data)

Boot, Peter. "A Pretty Sublime Mix of WTF and OMG'. Four Explorations into the Practice of Evaluation on Online Book Reviewing Platforms." *Journal of Cultural Analytics*, vol. 7, no. 2, Feb. 2023, <https://doi.org/10.22148/001c.68086>.

- **Keywords:** corpus analysis, text analysis, NLP
- **Summary:** The authors study book reviews using "corpus-based methodology" enabled by Sketch Engine.
- **Method:** Sketch Engine is a collocation tool that not only takes into account the adjacency of words but also their parts of speech and the grammatical relations between words. The author poses several questions in relation to book reviews, such as: to what extent do reviewers pay attention to plot, what is the role of the body in book reviews. Then they proceed to answer them by manipulating Sketch Engine, mainly by taking advantage of the parts of speech of words to find other words that are relevant to / modify / are modified by the former word.
- **Relevance:** This article is really helpful for demonstrating how to use Sketch Engine, an NLP tool, to answer questions that pertain to the attitudes, meanings, preferences, and habits behind a given text, which I think is pretty relevant.

Blankenship, Avery. "What We Didn't Know a Recipe Could Be: Political Commentary, Machine Learning Models, and the Fluidity of Form in Nineteenth-Century Newspaper Recipes." *Journal of Cultural Analytics*, vol. 9, no. 1, Apr. 2024, <https://doi.org/10.22148/001c.115371>.

- **Keywords:** machine learning, discussion about machine learning as method, machine learning models
- **Summary:** The author builds a machine learning model that can identify nineteenth century recipe forms/genres to argue that recipes are a flexible genre that gives space to political commentary and blurs the line between the public and the domestic. The author also argues that ML can be used to study form and genre as they focus on context and semantic relationships - rather than the exact content of what they are saying.
- **Method:** the author uses word embedding models and document vectors to train a machine learning model that can identify recipes in newspaper data. The author first trains the model (using Doc2vec instead of Word2vec with 19th century cookbook recipes, creating a vector (?) of these recipes that enable the

machine to identify new input as either recipe or not recipe based on how the new document is situated in relation to the existing matrix/vector (?). The author then gave more **newspaper** recipes to the model to increase the accuracy rate. The model was able to flag several documents that at first glance doesn't resemble a recipe but, as the author argues by reading them individually, actually transforms and borrows recipe forms to express socially and politically important messages. The author argues that embedding models can be used to study the genre/form/semantics of large data sets.

- **Relevance:** the author of the article spends a good chunk of the article explaining her approach to machine learning and how it fits with the work of digital humanists. The author provides helpful perspectives: they suggest that the goal is not to evaluate the efficacy of a given technology when used to analyze a corpus / archive, but to use technology as a perspective, where the researcher can identify gaps and disruptions - such as when the technology fails to meet expectations or when it makes decisions surprising to the researcher. I thought this was a very helpful perspective and appreciate how the author makes this point in reference to other digital humanists.

Leontyeva, Xenia, et al. "Gender (Im)Balance in the Russian Cinema: On the Screen and behind the Camera." *Journal of Cultural Analytics*, vol. 9, no. 1, May 2024, <https://doi.org/10.22148/001c.116142>.

- **Keywords:** gender bias, gender stereotype, representation of gender, statistical analysis, database markup
- **Summary:** The author modifies the Bechdel-Wallace test (which tests whether or not women are adequately represented in movies according to whether or not they have a non-man-related dialogue with another female character) to analyze the representation of female characters in contemporary Russian film. The test is modified to take into account not only whether or not female characters engage in dialogue, but also the proportion of dialogues among female characters. They use a marked-up database of movie dialogues from over 200 movies and uses manual markup to identify whether dialogues qualify

and the contexts of dialogues, then model the data statistically to calculate whether each movie satisfies the test. They also model the dialogue data in relation to a network analysis of the crew of the film to study gender representation on screen and off screen.

- **Method:** manually markup movie dialogues (text) in the database and analyze the data
- **Relevance:** While the researchers don't use NLP (instead using volunteers who manually markup and categorize the movie dialogues), I found the method and its modification of the BW test interesting. As the researchers suggest, instead of testing a binary yes or no regarding whether the movie fails or passes the test, they turn that binary into a proportion. While this is not about the "binary" of gender representation, and the researchers don't really acknowledge non-binary definitions of gender, I find the word choice intriguing - how would a non-binary and queer analysis of the BW test look like? How would the data be modeled in this case?

Gilkison, Aaron, and Maciej Kurzynski. "Vectors of Violence: Legitimation and Distribution of State Power in the People's Liberation Army Daily (Jiefangjun Bao), 1956-1989." *Journal of Cultural Analytics*, vol. 9, no. 1, May 2024, <https://doi.org/10.22148/001c.115481>.

- **Keywords:** NLP, vector embedding, topic modeling, text mining, representation, rhetoric, political rhetoric
- **Summary:** the article uses various text modeling and NLP techniques to analyze the People's Liberation Army Daily (a state-sponsored newspaper of the communist party) and study discourses of violence and militarization throughout 1956 - 1989. The first part of the article uses collocation to study three kinds of soldier figures (the "zhanshi," "junren," and "minbing") and the words/phrases most significantly associated with each. This is followed by a historical discussion of the changing meanings of the three figures. The second part of the article uses vector embedding and topic modeling to study violent language / language of violence throughout the publication. They identify changes in the degree of violence within the articles in relation to local and global conflicts, and

they use topic modeling to specify what these conflicts are and their significance. Additionally, they read the degrees of violence as varying rhetoric that responds to local and global crises of the party's legitimacy.

- **Method:**

- The first part of the article uses collocation to identify the top five words/phrases associated with each of the three figures of the soldier. The researchers don't really go in-depth to explicate the result but rely on it to explain historical changes in the significance of the three soldier figures.
- In the second part, their first step was to consistently identify articles that they will consider "violent" - they do this by querying how frequently violent terms selected by the researcher appear. Then, they select 100 "violent" articles from each year and transform them into vector representations (embeddings) with the help of a large encoder model. Then, these representations were averaged to create multiple "vectors of violence" using k-means clustering (basically finding multiple centers within the matrix of vectors to represent several averages of the content of violent articles).
- With the violent articles and their embeddings, they compute how close the articles from each year are to the vectors of violence, therefore generating a timeline that shows essentially how violent each year's articles are.
- Finally, they use topic modeling to identify which topics are the most concentrated in each year, and supplement it with historical and cultural analysis of the affective landscape/rhetoric of violence at each historical moment.

- **Relevance:** the quantitative and qualitative methods of the article seem very relevant. They use a combination of NLP / text analysis methods to understand the rhetoric behind an archive as it relates to the historical context. I like how the article links NLP techniques with theoretical discourses on violence, so that it doesn't feel like they are using these quant methods just for the sake of using them. They make claims such as: "topic modeling is a way of partitioning the textual space and aligns with the distributional perspectives of Rancière and Agamben."

Grisot, Giulia, and Berenike Herrmann. "Examining the Representation of Landscape and Its Emotional Value in German-Swiss Fiction between 1840 and 1940." *Journal of Cultural Analytics*, vol. 8, no. 1, Aug. 2023, <https://doi.org/10.22148/001c.84475>.

- **Key words:** sentiment analysis, quantitative method, digital humanities, text labeling
- **Summary:** The authors use computational methods to detect and examine how different types of space are distributed and affectively encoded in German-Swiss literature. Using text mining and sentiment analysis techniques, they map the emotions surrounding the urban/rural divide in literature from the 20th century. Using the method outlined below, they found more emotional richness that is attached to descriptions of the Swiss rural landscape especially within the work of Swiss writers writing in the German language. The authors conclude that this shows these writers' are influenced by ideas of national identity, causing them to distinguish themselves from other national German writers.
- **Method:** " In order to do this, we first compiled a comprehensive dictionary of named and non-named spatial entities in the broad spatial categories RURAL and URBAN, and examined the presence of sentiment and emotions (valence and discrete emotions) and their 'strength' (arousal) in relation to these. We used current state-of-the-art sentiment lexicons for German available to the digital humanities community. Similarly to Heuser et al., we mapped the spatial entities and the sentiment lexicons onto our corpus, and focused on spans of +/-50 words around the detected entities, in order to examine the specific sentiment and emotions related to space."
 - Overall, the author seems to use computational analysis to support a claim that is already assumed, rather than solely using quantitative methods to prove a point.
- **Relevance:** I find that the method is helpful to know, namely, using valances (basically positive/negative values that represent the emotion) and the strength/intensity of it. However, as the authors acknowledge, such a model does not account for contradictory and co-present emotions like the sublime

Bowman, Mike. "Text-Mining Metadata: What Can Titles Tell Us of the History of Modern and Contemporary Art?" *Journal of Cultural Analytics*, vol. 8, no. 1, June 2023, <https://doi.org/10.22148/001c.74602>.

- **Keywords:** NLP, topic modelling, word cloud, data analysis
- **Summary:** The author uses topic modelling, parts-of-speech tagging and word counting to study a dataset of modern and contemporary art titles throughout the past century. The author argues that this method reveals the meaning invested in art titles and how people understood the kind of knowledge that art (ought to) engender. The author also argues that the trend revealed by the method is aligned with canonical art history
- **Method:**
 - The first part of the paper uses topic modeling to identify 3 topics that correspond to three art historical periods in the history of modern/contemporary art. After identifying the three topics, the author uses a word cloud and identifies the most common words in each topic and provides an art historical reading of this result. They also graph the topics' development across time to further argue that they show distinct periodization.
 - The second part uses parts of speech tagging to look at trends in the use of different parts of speech (e.g. older art titles used more nouns whereas contemporary art titles use verbs)
 - The third part uses word counting, basically counting the number of words used in titles to show that contemporary art in fact tends to use longer titles.
- **Relevance:** I think the article is interesting as it combines close reading, data analysis, and art historical analysis. But, in terms of the method, I find that it is basically using digital methods to confirm existing and accepted narratives without really offering anything new - nor does it talk about the limitations or advantages of statistical methods (except, of course, confirming existing scholarship).

Conroy, Melanie. "Quantifying the Gap: The Gender Gap in French Writers' Wikidata." *Journal of Cultural Analytics*, vol. 8, no. 2, May 2023, <https://doi.org/10.22148/001c.74068>.

Keywords: gender, non-binary gender identity, gender gap, representation, wikidata, data analysis, statistics

Summary: The article examines gender gaps in Wikidata databases of French literary history and the history of Francophone writing. Specifically, the article looks at the representation of women (and non-binary) writers and the extent of their representation versus male writers within literary history data in Wikidata.

Method: The article's approach is very straightforward and concerned with scale / percentage rather than representation. It basically answers the question "how many women writers appear in Wikidata?" It also considers the quality of the data on women writers - examining dimensions like how many Wikipedia page links are there within a given Wikidata item, and in how many languages do women writers' data appear. Additionally, the article notes that there are currently 2,000 people in Wikidata assigned the gender "other," and, because of the small size, any of her results are inconclusive.

Relevance: I appreciate that the article takes a direct stab at examining what it calls the "gender gap." The author's attention to the statistical presence of nonbinary gender in Wikidata prompted me to think about the relationship between database taxonomies and nonbinary gender (although the author doesn't really talk about this). While reading this, I also wondered if the author's approach is conducive at all in terms of studying nonbinary people, because the article essentially categorizes male/female/other and identifies the scarcity of nonbinary gender identities (which is true statistically in any other field), but doesn't really say anything significant about the bigger infrastructures behind this.

Arnold, Taylor, et al. "Understanding Peanuts and Schulzian Symmetry: Panel Detection, Caption Detection, and Gag Panels in 17,897 Comic Strips Through Distant Viewing." *Journal of Cultural Analytics*, vol. 8, no. 3, Sept. 2023, <https://doi.org/10.22148/001c.87560>.

- **Keywords:** Computer vision, distant reading
- **Summary:**
 - This article applies computer vision and other algorithmics techniques to understand a large dataset of comic strips from Charles Schulz's *Peanuts*. Through their computational analysis, they argue that there is a large scale symmetry and stability across time that contributes to the work's popularity - this conclusion would not have been without the distant viewing/reading technique they used.
 - The authors make the claim that large scale / distant viewing methods allow us to look at the formal and structural qualities of a given object and medium (comics, in their case). One key question that they ask is "how does the image-text relationship manifest not just in a single comic strip, but in the entirety of a comic strip's print run?"
- **Method:**
 - The article explains at length the methods that they use and how they modify previous algorithms from computer vision to transform it for their study. In simple terms, they built custom "rule-based" algorithms based on the qualities of their archive (e.g. the common visual elements in comics).
 - The methods that they use include panel detection (how long a comic panel is), caption detection (identifying the location of captions in panels), and comics paratext (using computer vision to exclude copyright text, signatures, dates, etc.).
- **Relevance:** I find this article compelling in arguing that computational and algorithmic methods can be used to study form/medium from a distant position. I'm especially interested in how they modify existing algorithms and create "rules" for their modified algorithm to read their research objects. What does this say about the bias/unbias of algorithm? And how can the use of algorithms to study certain objects then reveal the implicit functioning of algorithms and how it is used?

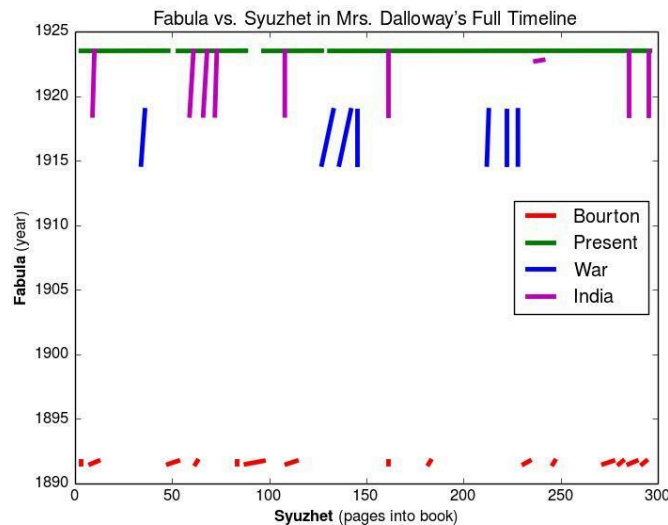
Ravina, Mark. "Algorithmic Maps and the Political Geography of Early-Modern Japan." *Journal of Cultural Analytics*, vol. 8, no. 3, Aug. 2023, <https://doi.org/10.22148/001c.84860>.

- **Keywords:** algorithm, mapping, digital humanities
- **Summary:** The article uses a specific kind of mapping algorithm to present how political leaders visualized their territories and the populations they ruled in early modern Japan.
- **Method:** The article relies on historical records of taxation and landholding to algorithmically create spatial entities on a map. The author first approximates significant geographical points from the historical record it examined (e.g. how villages were named in the record and where they are located, approximately). Then, he uses an algorithm called Voronoi tessellation to make those points into polygons - the polygons, as the author argues, are meant to approximate the political spaces embedded within the historical record.
- The KKT recorded the size, holder name, and village name for over 95,000 parcels in over 60,000 individual villages. This study combines the KKT data (as edited and digitized by the National Museum of Japanese History, or Rekihaku) with geolocation search engines to estimate the location of the villages.
- **Relevance:** I find the use of algorithms interesting in this study. For me, this article shows that algorithms can be used to show the implicit patterns within data, as the researchers do by using mapping algorithm on population and tax data to infer pseudo-borders during a period where spatial borders were constantly negotiated - this in turn show from historical taxation data the implicit views of those who regulated taxation and the creation of these data. The use of algorithms to show implicit “bias” or even epistemologies in objective data seems very relevant.

Yeager, Sean A. “Time Maps: Theory and Method.” *Journal of Cultural Analytics*, vol. 8, no. 3, Nov. 2023, <https://doi.org/10.22148/001c.87937>.

- **Keywords:** quantitative method, digital humanities
- **Summary:** The article creatively maps the fabula (the chronological events of a story) and syuzhet (how and where in the narrative the events are narrated) onto a Cartesian plane with fabula as the y-axis and syuzhet as the x-axis. The method is called a “time map” and visualizes the narratology of the story.

- Time map of Mrs. Dalloway:



- **Method:** going through the texts/films that the author studies and plotting elements on a
- **Relevance:**
 - It includes a literature review section where the author discusses the possibility of using automation and NLP/Machine Learning to understand the syuzhet, or how a story is told. Though the conclusion based on the author's own experience is negative, I find the discussion particularly helpful in thinking about the ability of LLMs/NLP to parse complex elements such as time.. Here are some of the articles cited:
 - Gregory Yuney, Ted Underwood, David Mimno's "Computational Prediction of Elapsed Narrative Time"
 - Underwood, Ted. "Using GPT-4 to Measure the Passage of Time in Fiction." *Using Large Digital Libraries to Advance Literary History*, 19 Mar. 2023.

Underwood, Ted. "Using GPT-4 to Measure the Passage of Time in Fiction." *Using Large Digital Libraries to Advance Literary History*, 19 Mar. 2023.

(<https://tedunderwood.com/2023/03/19/using-gpt-4-to-measure-the-passage-of-time-in-fiction/> not in Cultural Analytics)

- **Keywords:** LLMs, digital humanities, computational literary studies, ChatGPT
- **Summary:** the author uses chain of thought prompting to ask GPT4 to decide how much time has passed within a given literary passage and gets the r value between ChatGPT answers and the correct answer. He then compares the r value for GPT and the r value for agreement between two human readers from previous studies. He argues that ChatGPT can be used for content analysis but cannot be used to answer any questions about narrative time since “intersubjective agreement between human beings is still the only way to know that we have addressed those questions.”
- **Relevance:** I think it’s interesting how the author understands the limitation of ChatGPT by basically saying that it is impossible to measure certain things (like how much time has passed in a literary world) that only human intersubjective agreement can evaluate. I think it uses a humanist length to approach literary text and language, and I wonder what that means when the human is the benchmark of the limitations of technology or their value.

Thomas, Lindsay, and Abigail Droge. 2022. “The Humanities in Public: A Computational Analysis of US National and Campus Newspapers.” *Journal of Cultural Analytics* 7 (1): 36–80. <https://doi.org/10.22148/001c.32036>.

- **Keywords:** LLMs, modeling
- **Summary:** this article studies a database of US national and college newspapers to understand the discourse surrounding “the humanities” in public discussion.
- **Method:** The article uses various modeling/statistical methods to infer unique trends on how the “humanities” discourse is present in the dataset. For example, they use a **Wilcoxon rank sum test** to find the top most distinctive words as well as **topic modeling** to identify three major modes through which the humanities is present in their dataset (specifically, as “communicated,” “infrastructural,” or “debated”).
- **Relevance:** I think this article is very heavy on all kinds of topic modeling and keywords identification, and assigning their data into different categories and

conducting further analysis. It's interesting to see how the researchers interpret singular words (as topics or keywords) based on their statistical relationship to other words. I wonder if this might be relevant.

Berensmeyer, Ingo, and Sonja Trurnit. 2022. "Post-War British Women Writers and Their Cultural Impact: A Quantitative Approach." *Journal of Cultural Analytics* 7 (1): 81–107. <https://doi.org/10.22148/001c.33994>.

- **Keywords:** gender representation, quantitative analysis, feminist studies
- **Summary:** The authors conduct a quantitative study of the reception of women writers in post-war Britain using data from two magazines from the period. They compare the representation of women writers to their contemporary male writers, earlier British women writers, and canonical male writers.
- **Method:** the method is largely statistical, where they count the number of times women writers appear in the chosen material and basically compare the frequencies of different categories of writers being mentioned.
- **Relevance:** This is another article that studies gender disparities but doesn't really recognize non-binary gender. Even though I find queer approaches to something like literary history quite common, the same doesn't seem to be true when quantitative methods are used. I wonder if this is a limitation with quantitative methods?

Krasselt, Julia, and Philipp Dreesen. "Topic Models Indicate Textual Aboutness and Pragmatics: Valuation Practices in Islamophobic Discourse." *Journal of Cultural Analytics*, vol. 7, no. 2, Mar. 2024, <https://doi.org/10.22148/001c.92535>.

- **Keywords:** topic modeling, fear, islamophobia
- **Summary:** The authors use topic modeling to study a dataset of Islamophobic posts on a right wing German forum. Specifically, they look at posts where the Quran is quoted to trace the e/valuation of the Quran by the posters and how Islamophobic rhetoric is mobilized. Overall, they find that quotations of the Quran collocate strongly with language related to violence and terror.
- **Method:** The methods are really complicated topic modeling alongside some theories about linguistics and pragmatism. Essentially, the authors use topic

modeling but divides the topics that they get from the dataset into “aboutness” topics and “pragmatics,” where aboutness topics consist of words like nouns and adjectives that describe the object of the discussion (i.e. Islam) while pragmatics consists of action and hermeneutic words (like “ideology” or “read”). They also trace topic pathways that visualize how different topics intersect with each other. Overall, analyzing the “aboutness” and the “pragmatics” show how evaluative language blur the boundaries between thought and action, hence mobilizing fear and Islamophobia into violent action.

- **Relevance:** Although I didn’t quite get the topic modeling and the linguistics, I think the topic of this article is relevant, namely fear and rhetoric and how they not only formulate certain emotions but also mobilize action.